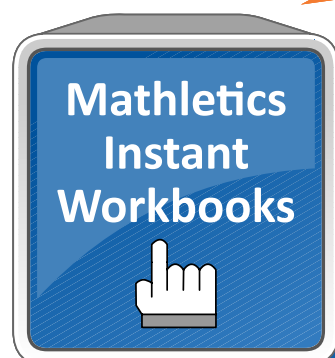


MATHLETICS

Percentages

Student Book - Series I-1

3.4% of 2560 = ____



Percentages

Student Book - Series I

Contents

Topics

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Author of The Topics and Topic Tests: AS Kalra

Percentages

Topic 1: Changing percentages to fractions

QUESTION 1 Express the following percentages as fractions in simplest form.

- | | | |
|----------------------|----------------------|-----------------------|
| a 50% = _____ | b 40% = _____ | c 30% = _____ |
| d 10% = _____ | e 20% = _____ | f 60% = _____ |
| g 70% = _____ | h 80% = _____ | i 25% = _____ |
| j 75% = _____ | k 85% = _____ | l 100% = _____ |
| m 17% = _____ | n 90% = _____ | o 95% = _____ |
| p 49% = _____ | q 19% = _____ | r 23% = _____ |

QUESTION 2 Express the following percentages as fractions in simplest form.

- | | | |
|------------------------|------------------------|------------------------|
| a 2.5% = _____ | b 8.2% = _____ | c 4.5% = _____ |
| d 36.5% = _____ | e 3.5% = _____ | f 27.5% = _____ |
| g 84.5% = _____ | h 40.2% = _____ | i 62.5% = _____ |
| j 0.5% = _____ | k 22.5% = _____ | l 0.8% = _____ |
| m 10.5% = _____ | n 8.5% = _____ | o 18.3% = _____ |
| p 9.25% = _____ | q 40.5% = _____ | r 65.5% = _____ |

QUESTION 3 Change the following percentages to fractions in simplest form.

- | | | |
|------------------------------------|------------------------------------|------------------------------------|
| a $\frac{1}{2}\%$ = _____ | b $\frac{1}{3}\%$ = _____ | c $\frac{1}{4}\%$ = _____ |
| d $\frac{1}{8}\%$ = _____ | e $22\frac{1}{2}\%$ = _____ | f $52\frac{1}{2}\%$ = _____ |
| g $5\frac{1}{2}\%$ = _____ | h $33\frac{1}{3}\%$ = _____ | i $66\frac{2}{3}\%$ = _____ |
| j $8\frac{1}{2}\%$ = _____ | k $5\frac{1}{2}\%$ = _____ | l $12\frac{1}{2}\%$ = _____ |
| m $10\frac{1}{2}\%$ = _____ | n $25\frac{1}{2}\%$ = _____ | o $35\frac{1}{2}\%$ = _____ |
| p $85\frac{1}{2}\%$ = _____ | q $15\frac{1}{2}\%$ = _____ | r $20\frac{1}{4}\%$ = _____ |

QUESTION 4 Convert the following to fractions in simplest form.

- | | | |
|------------------------------------|------------------------------------|-----------------------------------|
| a 38% = _____ | b 56% = _____ | c 62% = _____ |
| d $25\frac{1}{3}\%$ = _____ | e $12\frac{1}{4}\%$ = _____ | f $3\frac{1}{4}\%$ = _____ |
| g 9.5% = _____ | h 8.6% = _____ | i 4.8% = _____ |
| j 3.7% = _____ | k 6.9% = _____ | l 5.2% = _____ |
| m 42% = _____ | n 55% = _____ | o 72% = _____ |

Percentages

Topic 2: Changing fractions to percentages

QUESTION 1 Express the following fractions as percentages.

a $\frac{30}{100} =$ _____ b $\frac{50}{100} =$ _____ c $\frac{80}{100} =$ _____

d $\frac{40}{100} =$ _____ e $\frac{43}{100} =$ _____ f $\frac{84}{100} =$ _____

g $\frac{62}{100} =$ _____ h $\frac{39}{100} =$ _____ i $\frac{56}{100} =$ _____

j $\frac{25}{100} =$ _____ k $\frac{16}{100} =$ _____ l $\frac{72}{100} =$ _____

QUESTION 2 Express the following fractions as percentages.

a $\frac{5}{10} =$ _____ b $\frac{16}{10} =$ _____ c $\frac{7}{10} =$ _____

d $\frac{4}{5} =$ _____ e $\frac{6}{20} =$ _____ f $\frac{9}{50} =$ _____

g $\frac{9}{25} =$ _____ h $\frac{7}{25} =$ _____ i $\frac{7}{50} =$ _____

QUESTION 3 Change the following fractions to percentages.

a $\frac{3}{4} =$ _____ b $\frac{5}{80} =$ _____ c $\frac{13}{20} =$ _____

d $\frac{7}{60} =$ _____ e $\frac{2}{8} =$ _____ f $\frac{7}{40} =$ _____

g $\frac{9}{60} =$ _____ h $\frac{3}{5} =$ _____ i $\frac{14}{80} =$ _____

QUESTION 4 Express the following as percentages.

a $\frac{3}{8} =$ _____ b $\frac{5}{9} =$ _____ c $\frac{5}{8} =$ _____

d $\frac{7}{11} =$ _____ e $\frac{5}{12} =$ _____ f $\frac{1}{3} =$ _____

Percentages

Topic 3: Changing percentages to decimals

QUESTION 1 Express the following percentages as decimals.

- | | | |
|-----------------------|----------------------|----------------------|
| a 10% = _____ | b 80% = _____ | c 30% = _____ |
| d 50% = _____ | e 40% = _____ | f 60% = _____ |
| g 20% = _____ | h 70% = _____ | i 90% = _____ |
| j 100% = _____ | k 15% = _____ | l 65% = _____ |

QUESTION 2 Change the following percentages to decimals.

- | | | |
|----------------------|----------------------|----------------------|
| a 16% = _____ | b 19% = _____ | c 39% = _____ |
| d 12% = _____ | e 61% = _____ | f 56% = _____ |
| g 14% = _____ | h 73% = _____ | i 67% = _____ |
| j 98% = _____ | k 81% = _____ | l 78% = _____ |
| m 62% = _____ | n 43% = _____ | o 49% = _____ |

QUESTION 3 Convert the following percentages to decimals.

- | | | |
|-----------------------|-----------------------|-----------------------|
| a 5% = _____ | b 2% = _____ | c 8% = _____ |
| d 9% = _____ | e 6% = _____ | f 7% = _____ |
| g 3% = _____ | h 1% = _____ | i 4% = _____ |
| j 1.5% = _____ | k 2.5% = _____ | l 3.3% = _____ |
| m 2.1% = _____ | n 1.7% = _____ | o 4.9% = _____ |

QUESTION 4 Change the following percentages to decimals.

- | | | |
|------------------------|------------------------|------------------------|
| a 45.5% = _____ | b 89.2% = _____ | c 57.4% = _____ |
| d 63.9% = _____ | e 63.2% = _____ | f 36.5% = _____ |
| g 53.8% = _____ | h 92.7% = _____ | i 69.7% = _____ |
| j 23.7% = _____ | k 15.8% = _____ | l 21.3% = _____ |

QUESTION 5 Express the following percentages as decimals.

- | | | |
|------------------------------------|------------------------------------|------------------------------------|
| a $33\frac{1}{3}\%$ = _____ | b $\frac{5}{6}\%$ = _____ | c $\frac{3}{8}\%$ = _____ |
| d $1\frac{1}{2}\%$ = _____ | e $66\frac{2}{3}\%$ = _____ | f $12\frac{1}{2}\%$ = _____ |
| j $9\frac{1}{4}\%$ = _____ | k $20\frac{3}{4}\%$ = _____ | l $\frac{7}{9}\%$ = _____ |

Percentages

Topic 4: Changing decimals to percentages

QUESTION 1 Express the following decimals as percentages.

- | | | | | | |
|---|--------------|---|--------------|---|--------------|
| a | 0.15 = _____ | b | 0.25 = _____ | c | 0.35 = _____ |
| d | 0.95 = _____ | e | 0.14 = _____ | f | 0.85 = _____ |
| g | 0.18 = _____ | h | 0.55 = _____ | i | 0.17 = _____ |
| j | 0.45 = _____ | k | 0.67 = _____ | l | 0.65 = _____ |
| m | 0.91 = _____ | n | 0.75 = _____ | o | 0.72 = _____ |

QUESTION 2 Express the following decimals as percentages.

- | | | | | | |
|---|--------------|---|--------------|---|--------------|
| a | 0.80 = _____ | b | 0.60 = _____ | c | 0.1 = _____ |
| d | 0.50 = _____ | e | 0.90 = _____ | f | 0.2 = _____ |
| g | 0.70 = _____ | h | 0.4 = _____ | i | 0.3 = _____ |
| j | 0.01 = _____ | k | 0.02 = _____ | l | 0.05 = _____ |
| m | 0.04 = _____ | n | 0.06 = _____ | o | 0.03 = _____ |

QUESTION 3 Express the following decimals as percentages.

- | | | | | | |
|---|---------------|---|---------------|---|---------------|
| a | 0.123 = _____ | b | 0.625 = _____ | c | 0.537 = _____ |
| d | 0.189 = _____ | e | 0.427 = _____ | f | 0.139 = _____ |
| g | 0.835 = _____ | h | 0.567 = _____ | i | 0.683 = _____ |
| j | 0.129 = _____ | k | 0.816 = _____ | l | 0.927 = _____ |
| m | 0.687 = _____ | n | 0.257 = _____ | o | 0.369 = _____ |

QUESTION 4 Express the following decimals as percentages.

- | | | | | | |
|---|---------------|---|---------------|---|---------------|
| a | 0.001 = _____ | b | 0.910 = _____ | c | 0.006 = _____ |
| d | 0.007 = _____ | e | 0.002 = _____ | f | 0.009 = _____ |
| g | 0.810 = _____ | h | 0.037 = _____ | i | 0.021 = _____ |
| j | 0.004 = _____ | k | 0.005 = _____ | l | 0.003 = _____ |
| m | 0.057 = _____ | n | 0.089 = _____ | o | 0.065 = _____ |

QUESTION 5 Express the following decimals as percentages.

- | | | | | | |
|---|--------------|---|--------------|---|--------------|
| a | 1.20 = _____ | b | 2.50 = _____ | c | 3.60 = _____ |
| d | 2.56 = _____ | e | 1.31 = _____ | f | 3.57 = _____ |
| g | 8.25 = _____ | h | 3.41 = _____ | i | 5.63 = _____ |
| j | 2.84 = _____ | k | 5.34 = _____ | l | 8.13 = _____ |
| m | 5.27 = _____ | n | 8.03 = _____ | o | 2.06 = _____ |

Percentages

Topic 5: Finding the percentage of quantity

QUESTION 1 Evaluate the following.

- | | |
|----------------------|-----------------------|
| a 10% of 300 = _____ | b 20% of 5000 = _____ |
| c 30% of 900 = _____ | d 40% of 630 = _____ |
| e 50% of 580 = _____ | f 75% of 840 = _____ |
| g 25% of 370 = _____ | h 35% of 144 = _____ |
| i 5% of 210 = _____ | j 60% of 225 = _____ |
| k 70% of 890 = _____ | l 100% of 79 = _____ |
| m 45% of 125 = _____ | n 80% of 830 = _____ |
| o 85% of 369 = _____ | p 65% of 1000 = _____ |

QUESTION 2 Work out the following.

- | | |
|------------------------------------|-------------------------|
| a 1% of 48 = _____ | b 8.5% of 104 = _____ |
| c 2.5% of 36 = _____ | d 3.4% of 230 = _____ |
| e 6.9% of 83 = _____ | f 45.6% of 456 = _____ |
| g 4.3% of 55 = _____ | h 72.5% of 935 = _____ |
| i 62.5% of 890 = _____ | j 36.5% of 830 = _____ |
| k 12.5% of 88 = _____ | l 87.5% of 120 = _____ |
| m 37.5% of 120 = _____ | n 92.25% of 560 = _____ |
| o $33\frac{1}{3}\%$ of 180 = _____ | p 6.5% of 720 = _____ |

QUESTION 3 Calculate the following.

- | | |
|-----------------------------------|-----------------------------|
| a 17% of 500 = _____ | b 64% of \$350 = _____ |
| c 9% of 67 = _____ | d 18.5% of 366 days = _____ |
| e $5\frac{1}{2}\%$ of 840 = _____ | f 2.5% of \$80 = _____ |
| g 15% of \$120 = _____ | h 30% of 4 kg = _____ |
| i 40% of \$2 = _____ | j 400% of 30 = _____ |
| k 25% of 800 g = _____ | l 25% of \$812 = _____ |
| m 5% of \$1200 = _____ | n 142% of \$120 = _____ |
| o 90% of 50 = _____ | p 17% of 400 = _____ |

Percentages

Topic 6: Increasing or decreasing by a given percentage

QUESTION 1 Increase the following by the given percentage.

- | | |
|---------------------------------|--|
| a \$45 by 30% = _____ | b \$8000 by 15% = _____ |
| c \$80 by 45% = _____ | d \$3000 by 60% = _____ |
| e \$350 by 20% = _____ | f \$750 by 35% = _____ |
| g 900 g by 40% = _____ | h 50 minutes by 20% = _____ |
| i 400 kg by 60% = _____ | j 240 L by 50% = _____ |
| k 36 h by 100% = _____ | l 80 t by 75% = _____ |
| m \$648 by 25% = _____ | n \$320.80 by 16% = _____ |
| o \$199.95 by 5% = _____ | p \$149.90 by $12\frac{1}{2}\%$ = _____ |
| q 80 ha by 20% = _____ | r 600 cm by 70% = _____ |

QUESTION 2 Decrease the following by the given percentage.

- | | |
|----------------------------------|----------------------------------|
| a \$63 by 19% = _____ | b \$84 by 32% = _____ |
| c 4 m by 35% = _____ | d 260 L by 24% = _____ |
| e \$930 by 10% = _____ | f \$6354 by 20% = _____ |
| g \$134 by 25% = _____ | h \$575 by 30% = _____ |
| i \$928.50 by 15% = _____ | j \$243.80 by 50% = _____ |
| k 48 h by 25% = _____ | l \$840 by 70% = _____ |
| m \$365 by 15% = _____ | n \$960 by 45% = _____ |
| o \$96 by 40% = _____ | p \$60.50 by 80% = _____ |
| q \$940 by 28% = _____ | r \$620 by 35% = _____ |

QUESTION 3

- a** Bill bought a block of land for \$230 000. If the price of land rises by 80%, find the new value.

- b** Nathan earns \$1500 a week. Calculate his new weekly wage if he receives a pay rise of 20%.

- c** The price of a new car is \$30 000 and after 1 year it decreases by 20%. Find the new value of the car.

- d** A shirt priced at \$130 was sold at a discount of 25%. Find the sales price. _____

Percentages

Topic 7: Expressing quantities as percentages

QUESTION 1

- a Express 20 as a percentage of 200. _____
- b Express 12 as a percentage of 30. _____
- c Express 36 as a percentage of 63. _____
- d What percentage is 40 of 112? _____
- e What percentage is 28 of 77? _____

QUESTION 2 What percentage is:

- | | |
|------------------------|------------------------|
| a 20 of 80? _____ | b 60 of 240? _____ |
| c 14 of 35? _____ | d 15 of 60? _____ |
| e 28 of 56? _____ | f 14 of 42? _____ |
| g 19 of 152? _____ | h 81 of 90? _____ |
| i 110 of 440? _____ | j 10c of \$2? _____ |
| k 25c of \$3? _____ | l 63 cm of 1 m? _____ |
| m 16 h of 1 day? _____ | n 225 mL of 3 L? _____ |

QUESTION 3 In the following, what percentage is the first quantity of the second?

- | | |
|-------------------------|---------------------------|
| a 2 m, 16 m = _____ | b 8 s, 48 s = _____ |
| c 12 kg, 90 kg = _____ | d 6 L, 300 L = _____ |
| e 30 mL, 150 mL = _____ | f 125 g, 500 g = _____ |
| g 70 m, 210 m = _____ | h 80 min, 320 min = _____ |
| i 36 s, 2 min = _____ | j 30 cm, 3 m = _____ |
| k 18 h, 3 days = _____ | l \$2, 500 c = _____ |
| m 240 g, 6 kg = _____ | n 32 kg, 2 t = _____ |
| o 300 cm, 3.6 m = _____ | p 8 min, 2 h = _____ |

QUESTION 4 What percentage is:

- | | |
|-------------------------|-------------------------------------|
| a 50 c of \$6? _____ | b \$2.50 of \$10? _____ |
| c 28 min of 3 h? _____ | d 750 mL of 6 L? _____ |
| e 650 m of 4 km? _____ | f \$3.50 of \$4? _____ |
| g 70 c of \$3.50? _____ | h \$5 of 150 c? _____ |
| i 12 mm of 60 cm? _____ | j \$7.50 of \$30? _____ |
| k 45 cm of 81 cm? _____ | l 15 s of 50 s? _____ |
| m 600 g of 8 kg _____ | n 65 min of $4\frac{1}{3}$ h? _____ |

Percentages

Topic 8: Unitary method and percentages

QUESTION 1

- a If 50% of a number is 32, what is the number? _____
- b Find the total cost if 60% of it is \$90. _____
- c If 18% of a number is 9, what is the number? _____
- d If 150% of a number is 60, what is the number? _____

QUESTION 2 Use the unitary method to solve the following.

- a 20% of a number is 32, what is the number? _____
- b 10% of a number is 63, what is the number? _____
- c 15% of a number is 360, find the number. _____
- d 48% of a number is 336, find the number. _____

QUESTION 3 Find 100% of the number if a part of it is given.

- | | |
|----------------------------|--|
| a 40% is 80; 100% = _____ | b 20% is 92; 100% = _____ |
| c 25% is 120; 100% = _____ | d 15% is 60; 100% = _____ |
| e 50% is 70; 100% = _____ | f 75% is 450; 100% = _____ |
| g 5% is 83; 100% = _____ | h 45% is 119; 100% = _____ |
| i 2% is 94; 100% = _____ | j 8% is 34; 100% = _____ |
| k 16% is 46; 100% = _____ | l $12\frac{1}{2}\%$ is 230; 100% = _____ |

QUESTION 4

- a If $33\frac{1}{3}\%$ of a number is 9, find the number. _____
- b John's income increased by 3%. If his income rose by \$645, find his previous income. _____
- c Alku receives a 15% increase in her weekly wage.
If the increase is \$75, find her new weekly wage. _____
- d In the first year, the price of a block of land increased by 20%.
If this increase amounted to \$28 000, what is the price now? _____
- e Hari spends 40% of his weekly income on rent.
If his rent is \$220, find his income. _____
- f When an amount is increased by 20% it becomes \$65.
What was the original amount? _____
- g The population of a small town decreases by 8% every year.
If this year's population is 4600, what was the population last year? _____

Percentages

Topic 9: Problem solving and percentages

- 1 The price of a stereo is \$1200. The discount is 30%. What is the discount? _____
- 2 Out of 630 students, 80% swim. How many swim? _____
- 3 Out of 120 students, 65% went to the camp. How many went to the camp? _____
- 4 Clare obtained 90 out of 120 marks in a mathematics test.
What percentage was this? _____
- 5 Decrease \$100 by 10% and then increase this result by 10%. _____
- 6 If 35% of a number is 70, find the number. _____
- 7 A blanket that cost a retailer \$56.50 sells for \$110.
What percentage profit is made? _____
- 8 A salesman sold goods worth \$35 000. Find the commission he gained on the sale if the first \$10 000 sales earns 7% and the amount sold above \$10 000 earns 9%. _____
- 9 Find the simple interest on \$2000 for 1 year at 9.5% per annum. _____
- 10 Find the simple interest on \$3560 for 3 years at $8\frac{1}{4}\%$ p.a. _____
- 11 Find 55% of \$1640 and add it to 45% of \$150. _____
- 12 In an examination, 90% of a class of 30 students passed. How many students failed? _____
- 13 Michael bought a second-hand car for \$8000 and spent \$2300 repairing it. If he sold the car for \$15 600, find:
 - a his gain as a percentage of the total cost price (including cost of repairs). _____
 - b his gain as a percentage of the selling price. _____
- 14 A tiler makes a mixture for a bathroom floor with 5 blue tiles and 4 grey tiles to each white tile. Express the composition of the floor as percentages. Also find the number of blue tiles used in a floor of 1690 tiles.

- 15 What is the cost of a pair of shoes whose full price is marked at \$299.50, if a discount of $15\frac{1}{4}\%$ is allowed?

Percentages

Topic Test

PART A

Instructions

This part consists of 12 multiple-choice questions

Each question is worth 1 mark

Fill in only ONE CIRCLE for each question

Calculators are NOT allowed

Time allowed: 15 minutes

Total marks = 12

						Marks
1	16 marks out of 64 is what as a percentage?					
	(A) 20%	(B) 24%	(C) 25%	(D) 30%	1	
2	An item is bought for \$80 and sold for \$120. Find the percentage profit on the cost price.					
	(A) 40%	(B) 30%	(C) 60%	(D) 50%	1	
3	18% of a number is 27. The number is					
	(A) 150	(B) 200	(C) 250	(D) 300	1	
4	As a decimal $4\frac{1}{2}\%$, equals					
	(A) 0.42	(B) 0.45	(C) 0.042	(D) 0.045	1	
5	Ted pays \$60 after receiving a discount of 20%. The normal price equals					
	(A) \$63	(B) \$72	(C) \$75	(D) \$80	1	
6	0.3% equals					
	(A) $\frac{1}{3}$	(B) $\frac{3}{10}$	(C) $\frac{3}{100}$	(D) $\frac{3}{1000}$	1	
7	At a roll call, there were 4 students absent and 21 present. What percentage of the roll class was absent?					
	(A) 16%	(B) 19%	(C) 25%	(D) 84%	1	
8	A car was bought for \$6000 and sold at a profit of 30%. What was its selling price?					
	(A) \$1800	(B) \$4200	(C) \$6200	(D) \$7800	1	
9	The price of a tyre is reduced from \$80 to \$72. What is the percentage discount?					
	(A) 8%	(B) 10%	(C) 11%	(D) 90%	1	
10	What percentage is \$3 of \$20?					
	(A) 10%	(B) 12%	(C) 15%	(D) 18%	1	
11	If 120 is 20% of a number, then the number is					
	(A) 24	(B) 96	(C) 480	(D) 600	1	
12	$p\%$ of \$ q is					
	(A) \$ pq	(B) \$ $(\frac{p+q}{100})$	(C) \$ $100pq$	(D) \$ $(\frac{pq}{100})$	1	

Total marks achieved for PART A



Percentages

Topic Test

PART B

Instructions This part consists of 15 questions
Each question is worth 1 mark
Write answers in the answers-only column

Time allowed: 20 minutes

Total marks = 15

Questions	Answers only	Marks
1 Find 37% of \$300.		1
2 Increase 260 kg by 10% and then decrease the result by 10%.		1
3 Write $\frac{7}{12}$ as a percentage correct to two decimal places.		1
4 If 15% of an amount is \$140, what is the total amount?		1
5 After 25% of a bill has been paid, \$80 still remains to be paid. How much was the bill?		1
6 A concrete mixture contains one part cement, two parts sand and three parts gravel. What percentage of the mixture is sand?		1
7 Express 700% as a fraction.		1
8 Find the simple interest on \$800 at 9% p.a. for 1 year.		1
9 Express 52 as a percentage of 416.		1
10 Find 100%, given that 40% is 98.		1
11 David pays 28% of his income in tax. If he pays \$14 200 income tax, what is his income?		1
12 A pen is bought for \$69.50 and sold for \$89.50. What is the gain per cent?		1
13 Find 70% of \$8640 and decrease this amount by 15%.		1
14 \$4800 is invested in a fund giving 12% interest per year. How much interest is received for 5 years?		1
15 The price of a television set is reduced by 18% to \$1722. What was the original price?		1

Total marks achieved for PART B

15